

OOPS ASSIGNMENT

Question1.

Code :

```
#include<iostream>

using namespace std;

struct Student{
    string name;
    int rollno;
    string degree;
    string hostel;
    float CGPA;

    void addDetails(){
        cout<<"Enter Name : "<<endl;
        cin>> name;
        cout<<"Enter roll no. : "<<endl;
        cin>> rollno;
        cout<<"Enter degree : "<<endl;
        cin>> degree;
        cout<<"Enter Hostel : "<<endl;
        cin>> hostel;
        cout<<"Enter Current Cgpa : "<<endl;
        cin>> CGPA;
    }

    void updateDetails(){
        cout<< "Enter new roll no. " <<endl;
        cin>> rollno;
        cout<< "Enter new degree : " <<endl;
```

```
        cin>> degree;
    }
    void updateCGPA(){
        cout<< "Enter new CGPA : "<<endl;
        cin>> CGPA;
    }
    void updateHostel(){
        cout<< "Enter new Hostel : "<<endl;
        cin>> hostel;
    }
    void DisplayDetails(){
        cout<< "-----Student Details----- "<<endl;
        cout<< "Name : "<< name <<endl;
        cout<< "Roll no. "<< rollno << endl;
        cout<< "Degree : "<< degree << endl;
        cout<< "Hostel : "<< hostel <<endl;
        cout<< "Current CGPA : "<< CGPA <<endl;

    }
};

int main(){
    int n;
    cout<< "Enter number of students : "<<endl;
    cin>> n;
    Student s[n];
    for(int i = 0;i<n;i++){
        s[i].addDetails();
        s[i].DisplayDetails();
        s[i].updateCGPA();
        cout <<"Updated CGPA : "<< endl;
```

```
s[i].DisplayDetails();  
s[i].updateHostel();  
cout <<"Updated Hostel :"<< endl;  
  
s[i].DisplayDetails();  
}  
return 0;  
  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Enter number of students :
2
Enter Name :
Prateek
Enter roll no. :
1024150222
Enter degree :
biotech
Enter Hostel :
B
Enter Current Cgpa :
7.5
-----Student Details-----
Name : Prateek
Roll no. 1024150222
Degree : biotech
Hostel : B
Current CGPA : 7.5
Enter new CGPA :
8
Updated CGPA :
-----Student Details-----
Name : Prateek
Roll no. 1024150222
Degree : biotech
Hostel : B
Current CGPA : 8
Enter new Hostel :
D
Updated Hostel :
-----Student Details-----
Name : Prateek
Roll no. 1024150222
Degree : biotech
Hostel : D
Current CGPA : 8
Enter Name :
Panda
Enter roll no. :
1024150213
Enter degree :
enc
Enter Hostel :
B
Enter Current Cgpa :
7.2
-----Student Details-----
Name : Panda
Roll no. 1024150213
Degree : enc
Hostel : B
Current CGPA : 7.2
Enter new CGPA :
10
Updated CGPA :
-----Student Details-----
Name : Panda
Roll no. 1024150213
Degree : enc
Hostel : B
Current CGPA : 10
Enter new Hostel :
O
Updated Hostel :
-----Student Details-----
Name : Panda
Roll no. 1024150213
Degree : enc
Hostel : O
Current CGPA : 10
PS C:\Users\vaibh\OneDrive\Desktop\DSA>

● PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Hello, World!
○ PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question2.

Public members are accessible from outside the class, while **private members** are accessible only within the class and are used for data protection and encapsulation.

Code:

```
#include <iostream>

using namespace std;

class Student {
private:
    int rollNo;
    float marks;

    void calculateGrade() {
        if (marks >= 90)
            cout << "Grade: A\n";
        else if (marks >= 75)
            cout << "Grade: B\n";
        else
            cout << "Grade: C\n";
    }

public:
    void setData(int r, float m) {
        rollNo = r;
        marks = m;
    }

    void display() {
        cout << "Roll No: " << rollNo << endl;
        cout << "Marks: " << marks << endl;
        calculateGrade();
    }
};
```

```
    }  
};  
  
int main() {  
    Student* s1 = new Student;  
  
    s1->setData(101, 88.5);  
    s1->display();  
  
    return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }  
Roll No: 101  
Marks: 88.5  
Grade: B  
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question3.

Code:

```
#include <iostream>  
  
using namespace std;  
  
class Demo {  
private:  
  
    void privateFunction() {  
        cout << "This is a PRIVATE function\n";  
    }  
  
public:
```

```
void publicFunction() {  
    cout << "This is a PUBLIC function\n";  
    privateFunction();  
}  
};
```

```
int main() {  
    Demo obj;  
    obj.publicFunction();  
  
    return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }  
This is a PUBLIC function  
This is a PRIVATE function  
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question4.

Code:

```
#include <iostream>  
  
using namespace std;
```

```
class Rectangle {
```

```
private:
```

```
    int length;
```

```
    int breadth;
```

```
public:
```

```
    void getdata() {
```

```
        cout << "Enter length: ";
```

```
        cin >> length;
```

```
        cout << "Enter breadth: ";

        cin >> breadth;

    }

    void calculatearea() {

        int area = length * breadth;

        cout << "Area of Rectangle = " << area << endl;

    }

};

int main() {

    Rectangle r;

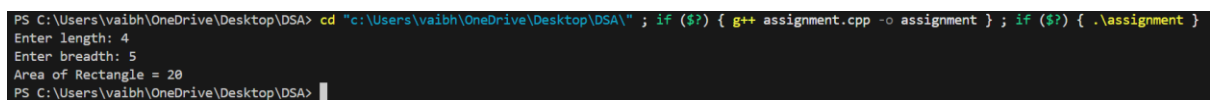
    r.getdata();

    r.calculatearea();

    return 0;

}
```

Output:



```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Enter length: 4
Enter breadth: 5
Area of Rectangle = 20
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question5.

Code:

```
#include<iostream>

using namespace std;

class Complex{

    private:

        float real;

        float imaginary;

    public :

        void setComplex(float r, float i){
```



```
    real = r;
    imaginary = i;
}

void displayComplex(){
    cout<< real;
    if (imaginary >= 0)
        cout<< "+" << imaginary << "i"<<endl;
    else
        cout<< "-" << -imaginary << "i"<<endl;
}

Complex addComplex(Complex c){
    Complex result;
    result.real = real + c.real;
    result.imaginary = imaginary + c.imaginary;
    return result;
}

};

int main(){
    Complex c1,c2,c3;
    float r,i;

    cout<< "Enter real and imaginary part of complex number : "<< endl;
    cin>> r>> i ;
    c1.setComplex(r,i);
    cout<< "Enter real and imaginary part of complex number : " <<endl;
    cin>> r>> i ;
    c2.setComplex(r,i);
    cout<< "First complex number : ";
    c1.displayComplex();
    cout<< "Second complex number : ";
```

```
c2.displayComplex();  
c3 = c1.addComplex(c2);  
cout<< "Sum of complex number : ";  
c3.displayComplex();  
return 0;  
  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }  
Enter real and imaginary part of complex number :  
12  
13  
Enter real and imaginary part of complex number :  
12 15  
First complex number : 12+13i  
Second complex number : 12+15i  
Sum of complex number : 24+28i  
PS C:\Users\vaibh\OneDrive\Desktop\DSA> █
```

Question 6:

(a)

Code:

```
#include <iostream>  
  
using namespace std;
```

```
class Test {
```

```
public:
```

```
    void show();  
};
```

```
void Test::show() {
```

```
    cout << "Function defined outside the class\n";  
}
```

```
int main() {
```

```
    Test t;  
    t.show();
```

```
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Function defined outside the class
PS C:\Users\vaibh\OneDrive\Desktop\DSA> █
```

(b)

Code:

```
#include <iostream>
```

```
using namespace std;
```

```
int x = 10;
```

```
int main() {
```

```
    int x = 20;
```

```
    cout << "Local x = " << x << endl;
```

```
    cout << "Global x = " << ::x << endl;
```

```
    return 0;
```

```
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }
Local x = 20
Global x = 10
PS C:\Users\vaibh\OneDrive\Desktop\DSA> █
```

(c)

Code:

```
#include <iostream>
```

```
using namespace std;
```

```
class Demo {
```

```
public:
```

```
static int count;  
};
```

```
int Demo::count = 5;
```

```
int main() {  
    cout << "Static variable = " << Demo::count << endl;  
    return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ assignment.cpp -o assignment } ; if ($?) { .\assignment }  
Static variable = 5  
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

(d)

Code:

```
#include <iostream>
```

```
int main() {  
    std::cout << "Enter a number: ";  
    int x;  
    std::cin >> x;  
    std::cout << "You entered: " << x << std::endl;  
    return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }  
Enter a number: 3  
You entered: 3  
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```

Question7.

Code:

```
#include <iostream>

using namespace std;

namespace First {

    int value = 10;

    void show() {
        cout << "First namespace value = " << value << endl;
    }
}

namespace Second {

    int value = 20;

    void show() {
        cout << "Second namespace value = " << value << endl;
    }
}

int main() {
    First::show();
    Second::show();

    cout << "First value = " << First::value << endl;
    cout << "Second value = " << Second::value << endl;

    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\DSA> cd "c:\Users\vaibh\OneDrive\Desktop\DSA\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
First namespace value = 10
Second namespace value = 20
First value = 10
Second value = 20
PS C:\Users\vaibh\OneDrive\Desktop\DSA>
```